ADVANCED INTERNET OPERATIONS RESEARCH IN INDIA

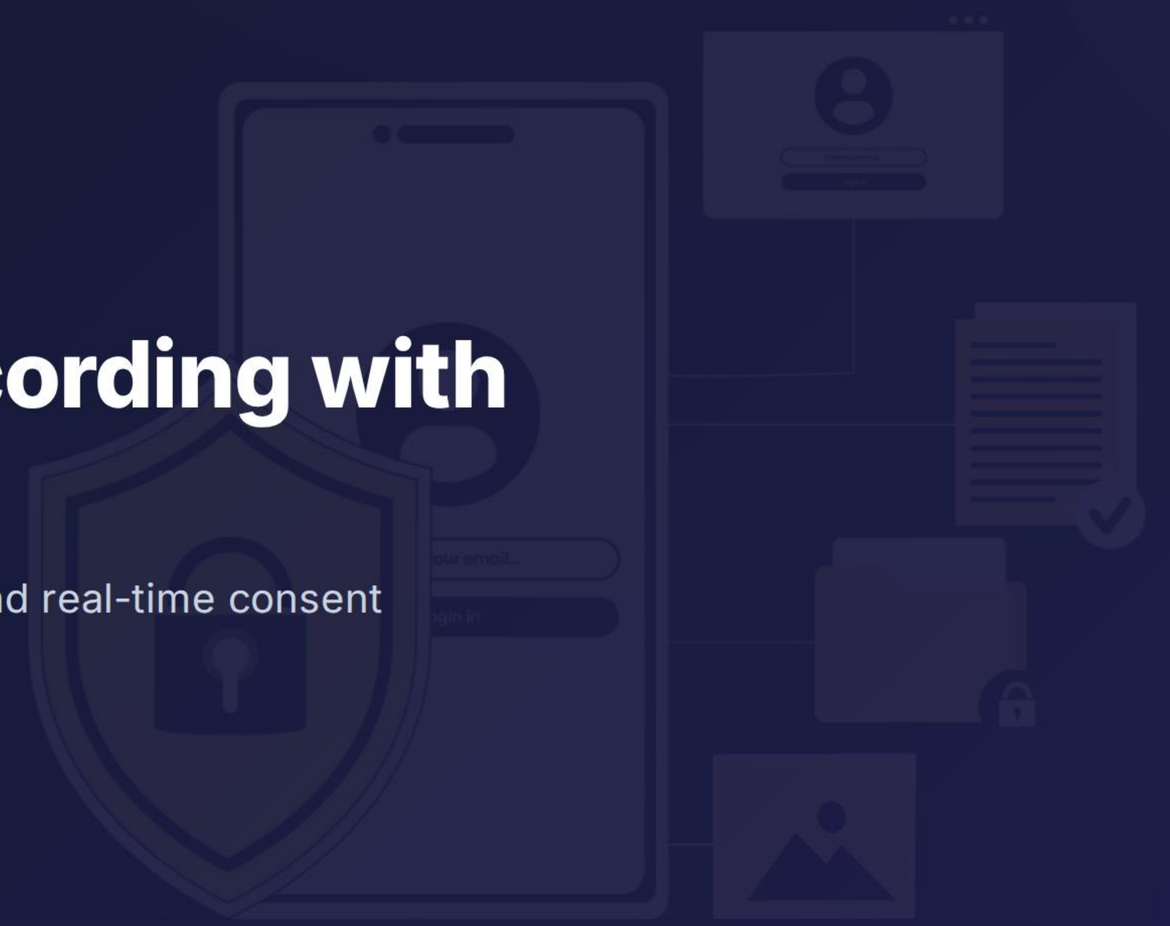
AIORI Hackathon 3

PROBLEM STATEMENT 002

Team Code: A3-PS002-TC314

Purpose-Based Consent Recording with Withdrawal Prototype

Dynamic scope authorization, immutable lifecycle audit logging, and real-time consent revocation mechanism.



TEAM NAME
Techslayer

TEAM LEADER
Kashish Gupta

TEAM MEMBER
Rashi Saraswat

FACULTY MENTOR
Vishnu Katara

October 2026

Problem Description

PS 002

Binary All-or-Nothing Consent

Monolithic terms force users into granting blanket access or forfeiting service access entirely.

Opaque Data Processing

Users lack real-time visibility into specific purpose scopes and downstream processing post-authorization.

Complex Withdrawal Flow

Existing systems lack standard dynamic revocation hooks, making withdrawal manual, slow, and unverified.



Regulatory & System Context



Regulatory Mandates

DPDP Act 2023 and GDPR Article 7 mandate explicit, purpose-linked consent and user-accessible, real-time withdrawal options.

Static DB Limitations

Legacy backends store binary consent flags without recording granular purpose state changes or timestamped lifecycles.

Auditability Need

Verifiable event logging is critical to demonstrate compliance, mitigate regulatory exposure, and preserve user trust.

Comparative Standards Review

RFC 6749 FRAMEWORK

OAuth 2.0 / OIDC Scopes

Provides fine-grained authorization scopes, but lacks native runtime revocation hooks for specific data scopes after token issuance.

W3C SPECIFICATION

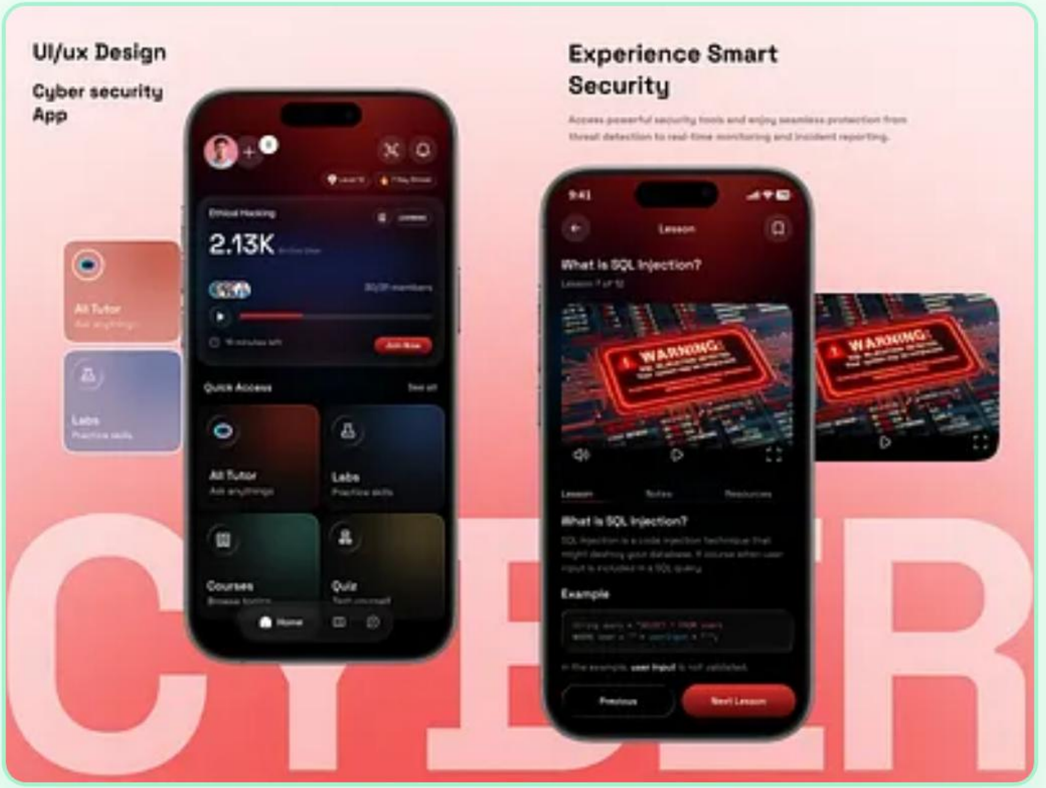
W3C Consent Receipt

Standardizes JSON receipts for consent records, but focuses on record representation rather than active revocation enforcement.

INDUSTRY WORKFLOWS

Legacy Web Banners

Rely on passive web UI banners or support emails, failing to deliver real-time data handling interception.



Technical Approach & Modules

PS 002

MODULE 01

Granular Scope Selector

Dashboard allowing users to individually toggle processing scopes (Core Service, Analytics, Marketing).

Dynamic UI

MODULE 02

Recording Engine

Captures every event (Grant, Revoke) with precise timestamps, scope IDs, and unique transaction hashes.

Event Engine

MODULE 03

Withdrawal Handler

Provides direct user revocation that updates scope states and blocks upstream processing requests instantly.

Instant Revoke

MODULE 04

Audit Log Store

Maintains a sequential, tamper-evident audit trail tracking consent state changes throughout the lifecycle.

Verifiable Logs

Implementation Scope: Purpose Selection, Event Logging & Real-Time Withdrawal Workflow

Validated Prototype

Feasibility & Capability Comparison

Architecture Dimension	Standard Consent Approach	Proposed Prototype Implementation
Consent Model	✕ Binary / All-or-Nothing Scope	✓ **Granular Purpose-Based Selection**
Withdrawal Process	✕ Manual / Support Email Request	✓ **Real-Time User Action / Single Toggle**
Auditability	✕ Static DB Entry	✓ **Timestamped State Transition Tracking**
Verification Scope	✕ Point-in-Time Static Flag	✓ **Dynamic Runtime State Interception**

Validation Status: Functional feasibility verified via prototype execution of consent grant, active verification, and instant withdrawal. Latency benchmarking remains *Future Work*.

System Architecture Diagram

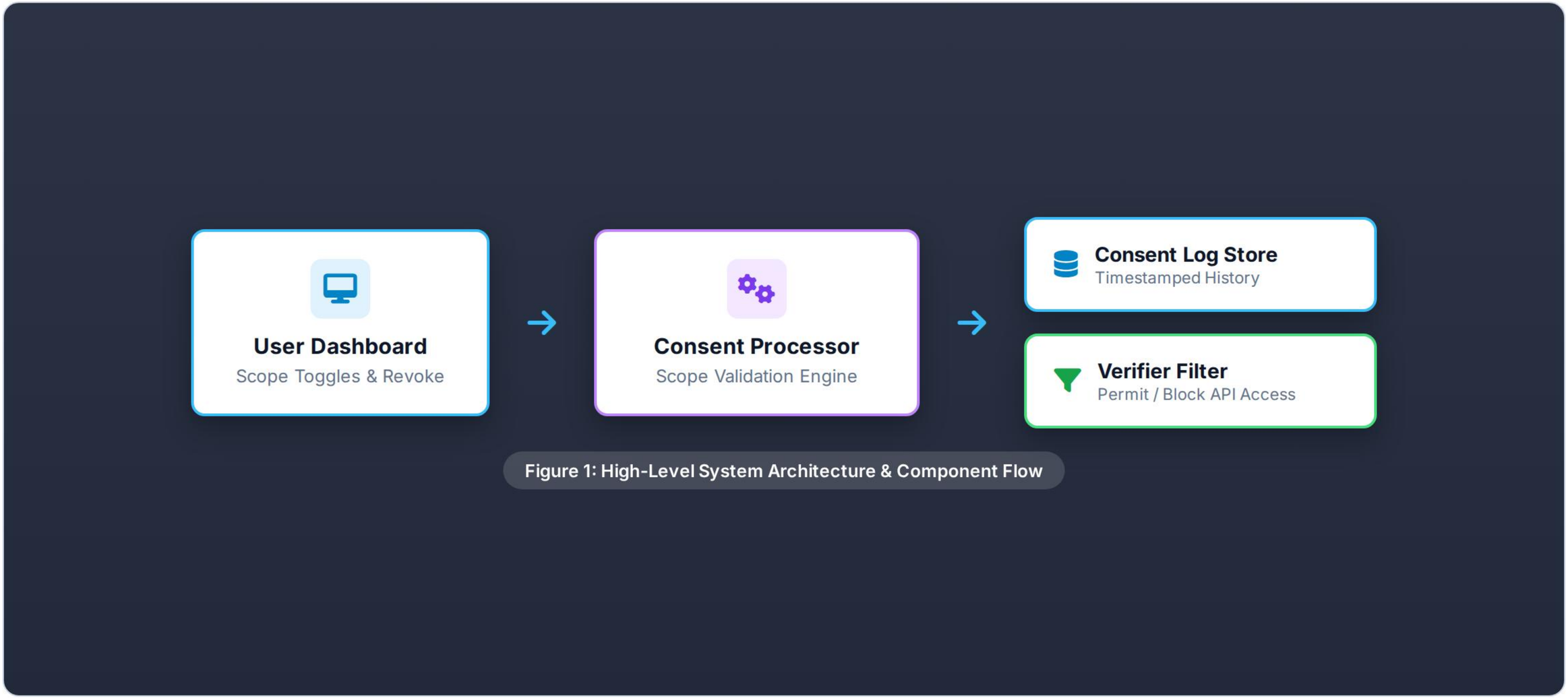


Figure 1: High-Level System Architecture & Component Flow

Consent Lifecycle Data Flow

PS 002

1. **Purpose Scope Selection**
   User selects active scopes and posts consent choices to the Consent Handler.

2. **Timestamped Event Logging**
   System persists record with user ID, scope IDs, timestamp, and status = ACTIVE.

3. **Access Verification Check**
   Upstream service queries Access Verifier to confirm active consent status before serving data.

4. **Revocation Execution**
   User triggers withdrawal → Status updates to REVOKED → Verifier blocks API with 403 Forbidden.



Milestones & Future Roadmap

PS 002

PHASE 1 (COMPLETED)

Core Prototype

Workflow modeling, granular purpose selection dashboard, and withdrawal execution handler development.

Completed

PHASE 2 (IN PROGRESS)

Validation & Refinement

Status verifier testing, audit log validation, and edge-case handling for simultaneous scope revocations.

In Progress

PHASE 3 (FUTURE SCOPE)

Scalability & Benchmarking

Controlled latency load benchmarking, tamper-evident cryptographic logging, and integration SDK creation.

Upcoming

References & Standards

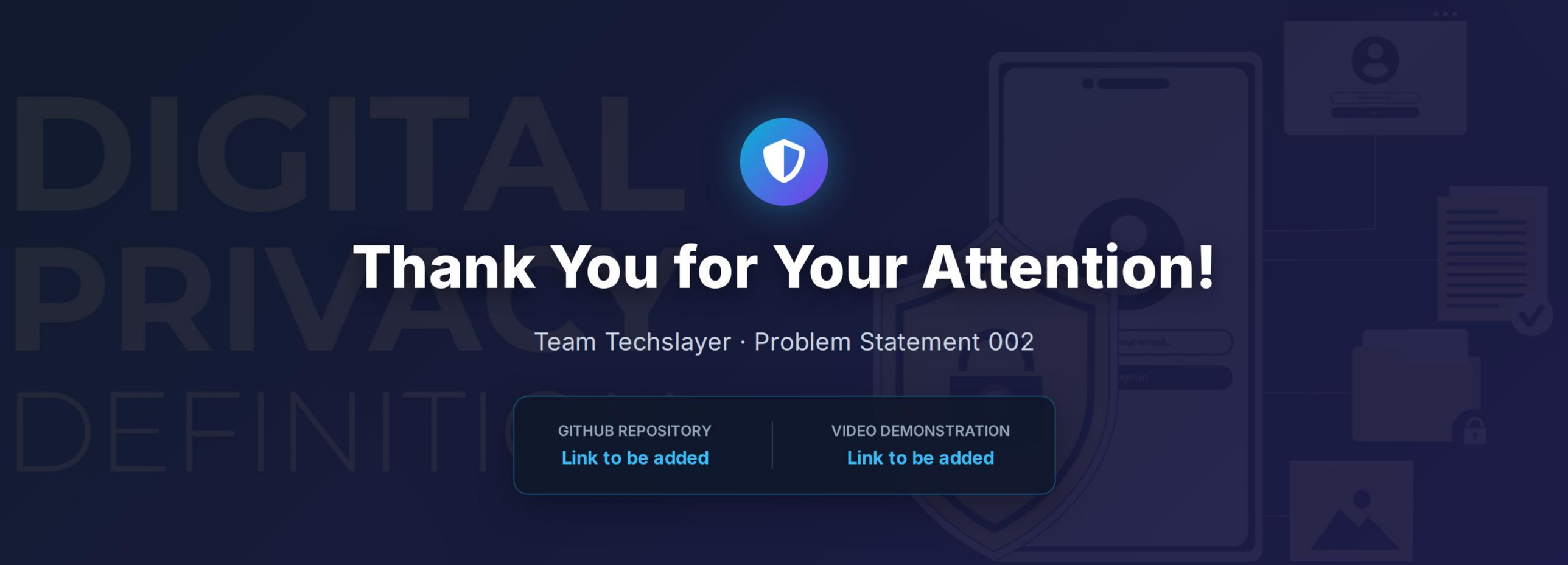
[1] **Digital Personal Data Protection (DPDP) Act**, Ministry of Electronics and Information Technology (MeitY), Government of India, 2023.

[2] **IETF RFC 6749**: Hardt, D., Ed., "The OAuth 2.0 Authorization Framework," Internet Engineering Task Force (IETF), October 2012.

[3] **ISO/IEC 27701:2019**: Security Techniques — Extension to ISO/IEC 27001 and ISO/IEC 27002 for Privacy Information Management.

AIORI HACKATHON 3

Team Code: A3-PS002-TC314



Thank You for Your Attention!

Team Techslayer · Problem Statement 002

GITHUB REPOSITORY	VIDEO DEMONSTRATION
Link to be added	Link to be added

Team Leader: Kashish Gupta

Team Member: Rashi Saraswat

Faculty Mentor: Vishnu Katara

Image Sources



https://www.enzuzo.com/hubfs/Mprgin_2_new_images_1920×1000_DB_14march2023_V1.png

Source: www.enzuzo.com



https://img.magnific.com/free-vector/network-infrastructure-server-room-topology-cloud-data-center-two-businessman_39422-429.jpg

Source: www.magnific.com



https://cdn.dribbble.com/userupload/49068559/file/06ae6a83f781e7e7d29a63b4122e5b03.png?crop=0x0-3200×2400&format=webp&resize=400×300&vertical=center

Source: dribbble.com



https://www.blockchain-council.org/_next/image/?url=%2Fblog-media%2Fposts%2Fp7kbsmrjjfow%2Fhow-data-immutability-works-in-blockchain.png&w=1920&q=75

Source: www.blockchain-council.org